

- Loki
- OpenTelemetry

With these tools, developers can utilise a combination of logs, metrics, and traces retrieved from multiple container systems to effectively monitor the health of the applications running in these different clusters.

Networking is another issue that needs to be addressed when working in multi-cloud environments. All workloads that are deployed across various cloud providers need to communicate with each other in a secure and efficient manner.

Service mesh technologies, such as Istio and Linkerd, help implement secure service-to-service communications and traffic management, as well as provide a means of performing distributed tracing and service discovery. By leveraging service meshes, organisations can apply networking policies uniformly across their various Kubernetes clusters.

Reference architecture for multi-cloud Kubernetes platforms

A multi-cloud Kubernetes environment generally has many levels or layers.

Kubernetes clusters: Kubernetes clusters can be managed using tools like Crossplane, Cluster API, etc. These help to automate the entire lifecycle of cloud infrastructure and cluster management.

Pipeline: Infrastructure and application configurations are stored in Git repositories so that they can be continually developed and managed through an automated process.

Service mesh layer: This allows secure service-to-service communication between clusters, like Istio and Linkerd.

Observability stack: Prometheus and Grafana provide observability/monitoring of workloads across multiple clusters.

With this architecture, an organisation can deploy workloads

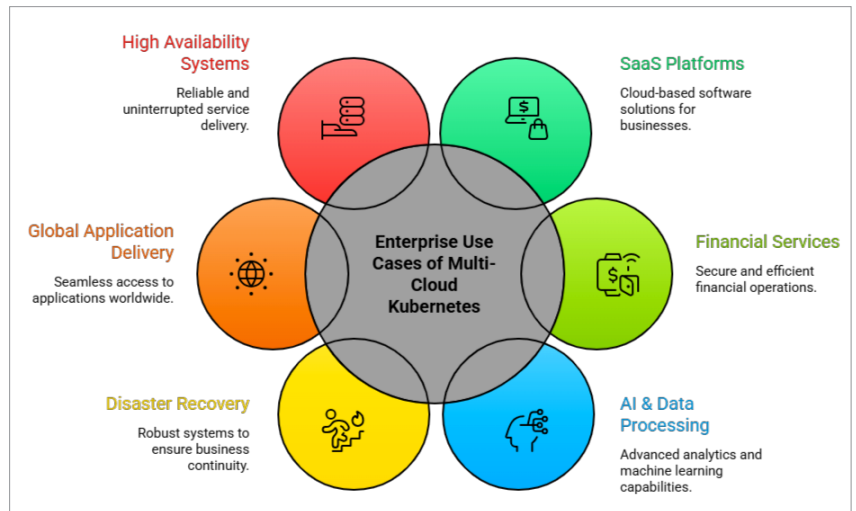


Figure 3: Use cases of multi-cloud Kubernetes

to multiple public clouds in a unified way, with a single point of control, management, and governance.

Real-world use cases of multi-cloud Kubernetes

Multi-cloud designs based on Kubernetes are being used by most businesses.

Software as a Service (SaaS):

Global SaaS companies are using multi-cloud to deliver their products more reliably. These architectures are often implemented by experienced .NET development companies that design scalable microservices and container-based applications running on Kubernetes clusters.

Financial services: Companies in finance use multiple cloud suppliers to meet the expectations of their regulators, so that they don't become too dependent on one vendor.

AI and data platform: Large data processing jobs run across multiple clouds to ensure that compute resources are always available for data processing.

Disaster recovery: Organisations create replicas of their production environments in a different cloud so that they can carry on with their business even when the primary service provider fails to provide the compute resources needed.

Building multi-cloud designs typically calls for considerable experience with DevOps best practices, as well as automating infrastructure. Therefore, most companies partner with experienced cloud or Kubernetes consultants to help design and create dynamic multi-cloud platforms.

Companies that offer Kubernetes consulting services (e.g., developing Kubernetes solutions), DevOps services (e.g., automating DevOps), and the development of cloud-native applications help businesses build production-ready environments and structured pipelines between their multi-cloud providers.

Challenges in managing multi-cloud Kubernetes

Although there are many advantages of using a multi-cloud Kubernetes management system, there are still a lot of challenges involved in managing one.

Network latency: Different cloud providers can result in additional delay when communicating between different workloads.

Security management: The identity and access control policies need to be the same across every cluster.

Cost management: Poor optimisation can increase the operational costs of running different infrastructures in different clouds.